

DBMS PROJECT REPORT

REQUIREMENT ANALYSIS AND CUSTOMER DATABASE MODULE

1. Introduction

The Requirement Analysis and Customer Database Module is designed to manage customer information in a structured database system. The module stores important customer details such as customer ID, customer name, email, phone number, address, and other required information. The database helps maintain customer records accurately and makes it easier to retrieve, update, and analyze customer information.

2. Requirement Analysis

Scenario

The system requires a customer database to maintain complete and organized customer information. The database should store customer details, ensure that important fields are not left empty, prevent duplicate email IDs, and allow customer records to be easily searched and analyzed.

The main requirements are:

1. Create a Customer Table.
2. Define appropriate columns and data types.
3. Establish a Primary Key for uniquely identifying each customer.
4. Apply suitable constraints to maintain data accuracy.
5. Prevent duplicate customer email IDs.
6. Make important customer fields NOT NULL.
7. Insert customer records into the database.
8. Retrieve and analyze customer information using SQL queries.

3. Customer Database Table

The Customer Table is used to store the details of customers.

Column	Purpose
Customer_ID	Unique identification of each customer
Customer_Name	Stores the customer's name
Email	Stores the customer's email address
Phone	Stores the customer's phone number
Address	Stores the customer's address

The table structure should be designed according to the requirements of the project.

```
SQL> CREATE TABLE CUSTOMER (CUSTOMER int PRIMARY KEY , CUSTOMER_NAME VARCHAR(100) NOT NULL, EMAIL VARCHAR(100) UNIQUE  
,PHONE VARCHAR(15) UNIQUE,ADDRESS VARCHAR (200),REGISTRATION_DATE DATE NOT NULL);  
  
Table created.
```

4. Primary Key

The Customer_ID is used as the primary key of the Customer Table. A primary key uniquely identifies every customer record in the database. It prevents two records from having the same customer ID.

5. Constraints

Constraints are used to maintain the accuracy and integrity of the customer data.

5.1 NOT NULL Constraint

Important fields such as customer name and email can be defined as NOT NULL so that these fields cannot be left empty.

5.2 UNIQUE Constraint

The Email column can be given a UNIQUE constraint so that the same email ID cannot be entered for multiple customers. This helps prevent duplicate customer records.

5.3 Primary Key Constraint

The Customer_ID column is defined as the primary key to uniquely identify each customer.

6. Insert Customer Values

After creating the Customer Table and applying the required constraints, customer records are inserted into the table.

The inserted data contains customer information such as:

- Customer ID
- Customer Name
- Email
- Phone Number
- Address

```

SQL> INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'));
Enter value for customer_id: 101
Enter value for customer_name: BECCA
Enter value for email: Becca@gmail.com
Enter value for phone: 1234567891
Enter value for address: thailand
Enter value for date: 2002-05-21
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('101','BECCA','Becca@gmail.com','1234567891','thailand',TO_DATE('2002-05-21','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 102
Enter value for customer_name: rebecca
Enter value for email: rebecca@gmail.com
Enter value for phone: 9875453643
Enter value for address: england
Enter value for date: 2008-02-22
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('102','rebecca','rebecca@gmail.com','9875453643','england',TO_DATE('2008-02-22','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 103
Enter value for customer_name: patricia
Enter value for email: pat@gmail.com
Enter value for phone: 9876543211
Enter value for address: british
Enter value for date: 2007-09-09
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('103','patricia','pat@gmail.com','9876543211','british',TO_DATE('2007-09-09','YYYY-MM-DD'))

1 row created.

```

```

SQL> /
Enter value for customer_id: 104
Enter value for customer_name: armstrong
Enter value for email: strong@gmail.com
Enter value for phone: 1234567814
Enter value for address: pattaya
Enter value for date: 2009-02-21
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('104','armstrong','strong@gmail.com','1234567814','pattaya',TO_DATE('2009-02-21','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 105
Enter value for customer_name: becbec
Enter value for email: becbec@gmail.com
Enter value for phone: 9876543121
Enter value for address: star
Enter value for date: 2004-07-22
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('105','becbec','becbec@gmail.com','9876543121','star',TO_DATE('2004-07-22','YYYY-MM-DD'))

1 row created.

```

7. Display Customer Records

SQL queries can be used to retrieve all customer records from the Customer Table. The displayed result helps verify that the customer information has been successfully stored in the database.

```
SQL> SELECT * FROM CUSTOMER;
```

CUSTOMER	CUSTOMER_NAM	EMAIL	PHONE	ADDRESS
REGISTRATI				
101	BECCA	Becca@gmail.com	1234567891	thailand
21-MAY-02				
102	rebecca	rebecca@gmail.com	9875453643	england
22-FEB-08				
103	patricia	pat@gmail.com	9876543211	british
09-SEP-07				
CUSTOMER	CUSTOMER_NAM	EMAIL	PHONE	ADDRESS
REGISTRATI				
104	armstrong	strong@gmail.com	1234567814	pattaya
21-FEB-09				
105	becbec	becbec@gmail.com	9876543121	star
22-JUL-04				

8. Customer Database Analysis

The stored customer information can be analyzed using SQL queries. The following reports can be generated from the Customer Table.

Report 1: Customer Details Report

Display:

- Customer ID
- Customer Name

- Email
- Phone Number
- Address

This report displays the complete information of customers stored in the database.

Report 2: Customer Search Report

The Customer Table can be searched using specific customer information.

For example, customers can be searched based on:

- Customer ID
- Customer Name
- Email
- Phone Number

This helps the system quickly retrieve a particular customer's information.

Report 3: Customer Analysis

The database can also be used to perform basic customer analysis.

Find:

- Total number of customers
- Customers based on specific criteria
- Customer records with available contact information
- Duplicate or invalid records, if any

SQL aggregate functions such as COUNT() can be used to calculate the total number of customers.

9. Data Integrity

The Customer Database Module maintains data integrity by using primary keys and constraints. The Primary Key ensures that each customer has a unique identification number. The UNIQUE constraint prevents duplicate email IDs. The NOT NULL constraint ensures that important customer information is entered. These constraints help maintain accurate and reliable customer records.

10. Conclusion

The Requirement Analysis and Customer Database Module provides a structured way to store and manage customer information. The project includes the creation of the Customer Table, defining primary keys, applying constraints, inserting customer records, retrieving customer information, and generating customer reports. The module demonstrates how a DBMS can be used to maintain customer data efficiently while ensuring data accuracy, uniqueness, and integrity.